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Calibration improves observer reliability in detecting periapical pathology on panoramic radiographs

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ISTORY

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Introduction

Radiography is an essential tool in many areas of dentistry, not the least in the field of endodontics. Endodontic inflammatory conditions, such as pulpitis and apical periodontitis, do not always exhibit any clinical signs or symptoms. Detection of carious and periapical lesions are both highly dependent on radiographic examination. Intra-oral radiography is the most commonly used method for caries and periapical diagnosis. Cone-beam computed tomography (CBCT) has recently become a useful adjunct in certain cases. Compared with intra-oral radiography, panoramic radiography (PR) is of limited value for endodontic diagnosis [1,2] or for detection of approximal caries [3]. Compared with CBCT, an even greater discrepancy is reported [4,5]. However, in the absence of an independent gold standard, the results should be interpreted with caution.

PR offers a rapid, simple means of providing a broad overview of both jaws and teeth and is likely to be well accepted by patients and/or study participants. It is therefore commonly used in epidemiological studies. Variables such as the number of remaining teeth, restorative therapy and endodontic treatments are accurately disclosed. Despite low sensitivity in detecting caries and periapical pathology, for larger epidemiological studies PR is considered to provide an adequate overview [1,2].

Any study reliant on interpretation of a radiograph, regardless of method, is subject to intra- and inter-observer variation. This issue was clearly demonstrated in the classic study by Goldman et al. [6]. Six observers were instructed to determine independently the periapical status of endodontically treated teeth on intra-oral radiographs: there was agreement in fewer than 50% of the cases. Subsequent studies have also reported on the issue of reliability in radiographic

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assessment of periapical pathology, both for conventional [7,8] and digital [9] intra-oral radiographs, and more recently also for CBCT [10]. Epidemiological and large-scale studies often involve multiple observers and variation in assessment between two or more observers can greatly influence the results. Uncertainties in the registration of independent variables will have implications for the reliability of any conclusions drawn about the material.

Strategies have been proposed for improving reliability in radiographic assessment. The positive effects of observer calibration have been reported in a number of studies in which periapical status was assessed on intra-oral radiographs [7,11,12]. However, to our knowledge, there are no such studies with respect to periapical pathology detected exclusively on PR. As PR is currently widely used in epidemiological studies, and this is likely to continue, e.g. for studies investigating associations between endodontic inflammatory disease and other conditions, the issue of reliability must be addressed. The aim of this study was to investigate whether calibration against a reference standard improves observer reliability when using PR to assess variables indicative of endodontic inflammatory disease, including calculation of DMFT-score and detection of root-filled teeth and periapical lesions.

Materials and methods

In accordance with the instructions for calibrating the periapical index (PAI) [13], the sample selected for this study comprised 100 PRs from a cohort previously described by Ryden

et al. [14]. In short, the cohort comprised clinical data and PRs from 797 patients who had suffered an initial myocardial infarction and 796 controls, matched for age, sex and postal code area. Dental examinations, as well as radiographic examinations with either film or digital PRs, were undertaken during 2010-2014 at dental clinics at, or near, 17 different Swedish hospitals.

Training and calibration procedures

The study layout is illustrated in Figure 1.

Initially, two experienced specialists, one endodontist (TK) and one radiologist (HL), served as reference examiners. Prior to the calibration process instructions were formulated for the radiographic assessment of the following variables; remaining teeth, root-filled teeth, teeth with periapical lesions and DMFT-score, calculated from assessment of decayed, missing and filled teeth.

The instructions were as follows: 'Remaining teeth' comprise, besides apparent ones, also root remnants which are at least partially connected to the surrounding bone, impacted teeth covered by bone (mucosa and follicle) by no more than 2/3 of the crown and also any deciduous teeth. 'Root-filled teeth' are any teeth in which the pulp cavity and/or root canals contain radiopaque material. 'Teeth with periapical lesions' are any teeth that present distinct periapical bone destruction, noticeably larger than a widened periodontal space, around at least one root. For 'DMFT-score'

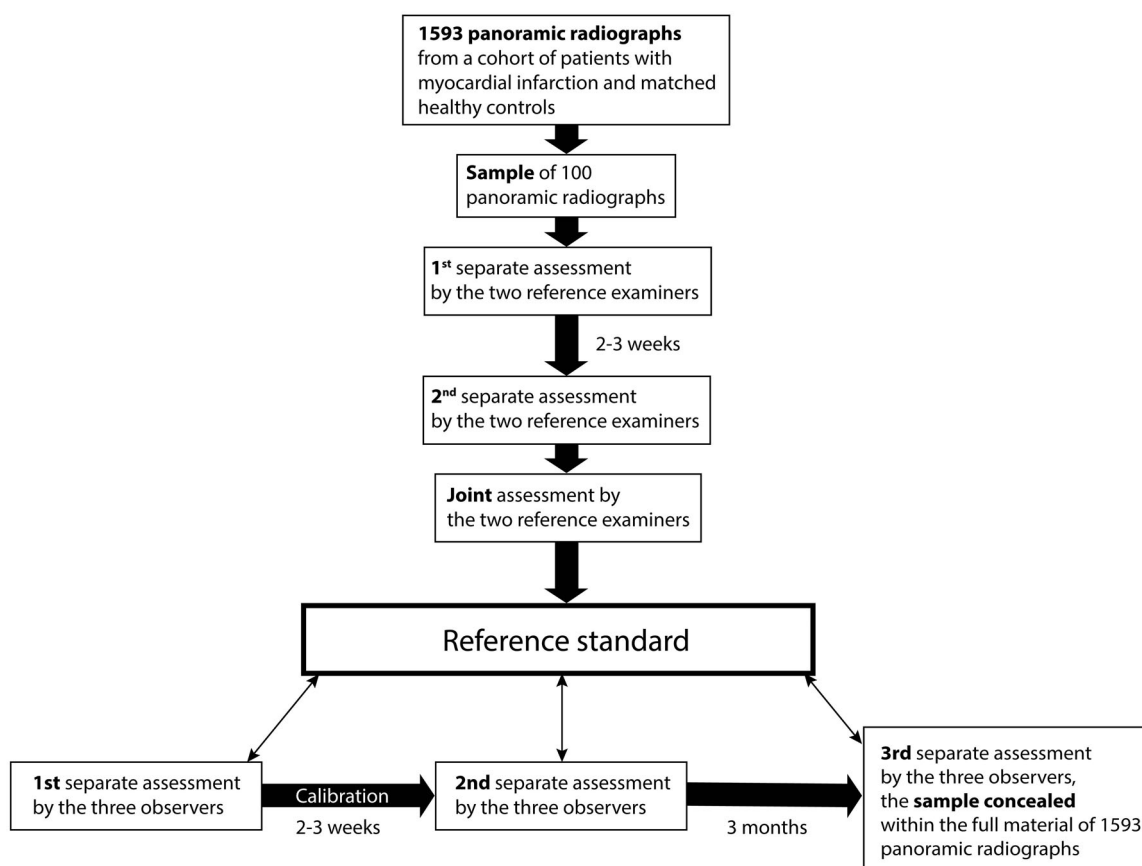


Figure 1. Flow chart of the study.

root remnants lacking restoration are registered as decayed teeth and dental implants are registered as missing teeth.

All PR assessments were made on the same high-resolution screen, in a dimly lit room. All images were assessed using the image-processing program ImageJ (U.S. National Institutes of Health, Bethesda, Maryland, USA, <https://imagej.nih.gov/ij/>), allowing adjustment of contrast and use of magnification tools. The two specialists assessed the sample independently and re-examined the material 3-4 weeks later, in order to determine intra-observer variation. The results were analysed and all cases with either intra- or inter-observer variations were subsequently subjected to a third, joint assessment, in which contradictory findings were discussed until consensus was reached. The results from the joint assessment then served as the reference standard for all future assessments of this sample.

Three observers, one general dentist (observer 1) and two fifth-year dental students (observers 2 and 3), were presented with written instructions for the radiographic assessment. Each observer made an initial, independent assessment of the sample. Thereafter the results were compared with the reference standard and the agreement was calculated. For calibration, each observer was individually presented with the results and given instructions on how to achieve closer agreement with the reference standard. Specifically, two observers (2 and 3) were directed to apply the criteria for registering periapical lesions more stringently, while one observer (1) was directed to lower slightly the threshold for reporting periapical lesions, to match the reference standard more closely. A second assessment of the sample followed and the results were again compared with the reference standard.

In order to evaluate the long-term effect of the calibration process, a third assessment was undertaken approximately three months later. This time the sample was randomly concealed in the complete material of 1593 PRs. When the entire material had been reassessed, agreement between the sample and the reference standard was again calculated.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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Conclusions

The study confirms that variables such as the number of remaining teeth, the number of root-filled teeth and the DMFT-score can be reliably assessed on PR. There was initially high variability in recording of teeth with periapical lesions. However, after calibration against a reference standard, agreement improved in all observers, resulting in increased reliability in detection of periapical lesions on PRs.

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